

Compositional gain model — parameter glossary

$$M(\mathbf{x}, t) =$$

Baseline gain
(no modulation)

1

+

Sustained
HP gain field

$g_{HP} G_{HP}(\mathbf{x}; \mu_{HP}, \sigma_{HP})$

+

Transient gain
at distractor onset

$g_D G_D(\mathbf{x}; \mu_D, \sigma_D) d(t)$

+

Transient gain
at target onset

$g_T G_T(\mathbf{x}; \mu_T, \sigma_T) \tau(t)$

Per VOXEL
PRF params

μ_{PRF}

σ_{PRF}

amp

baseline

Per ROI × subject
Gain-field params

g_{HP}

g_D

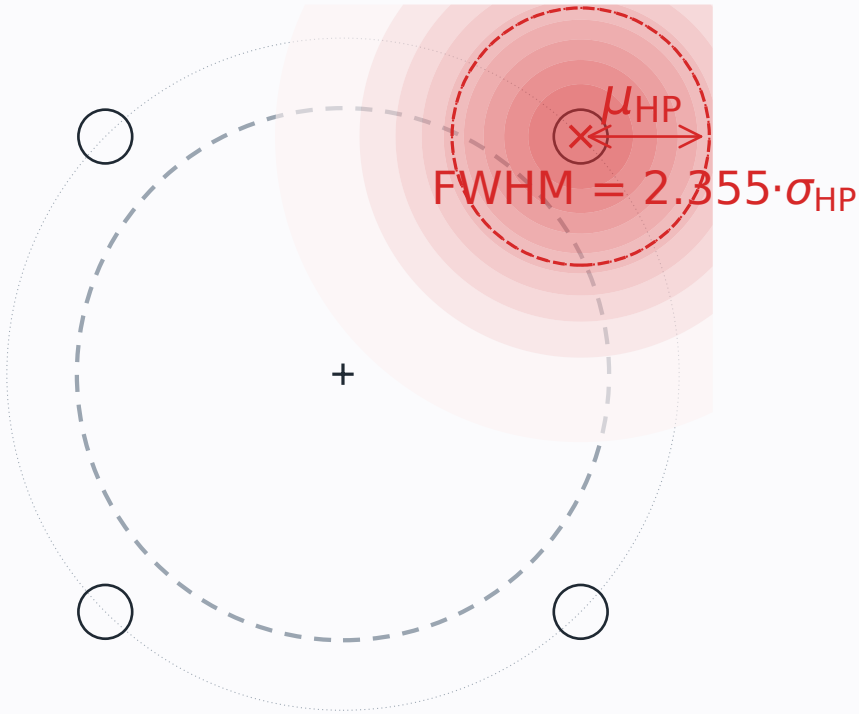
g_T

σ_{HP}

σ_{dyn}

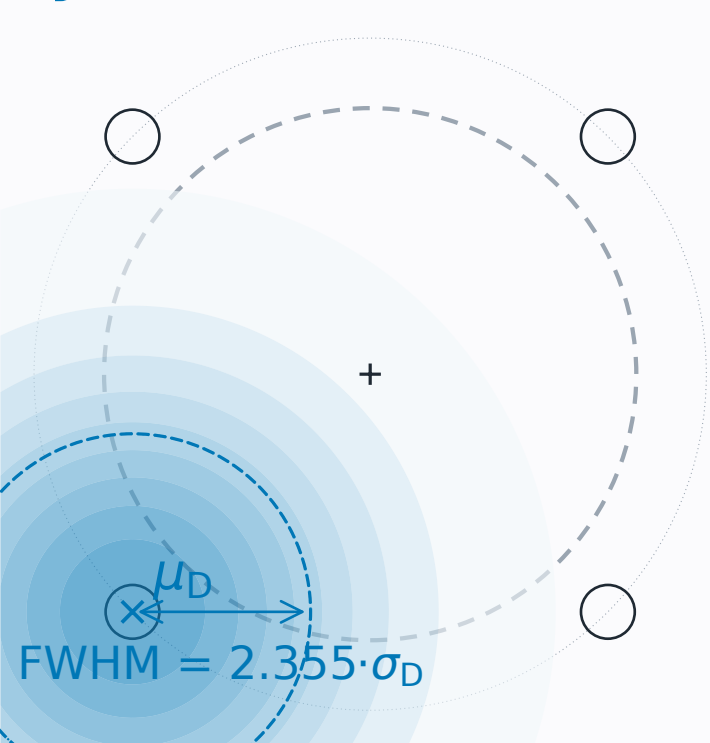
Fixed by paradigm
(trial events)
 $\mu_{HP/D/T}, d(t), \tau(t), S(\mathbf{x}, t)$

Sustained HP



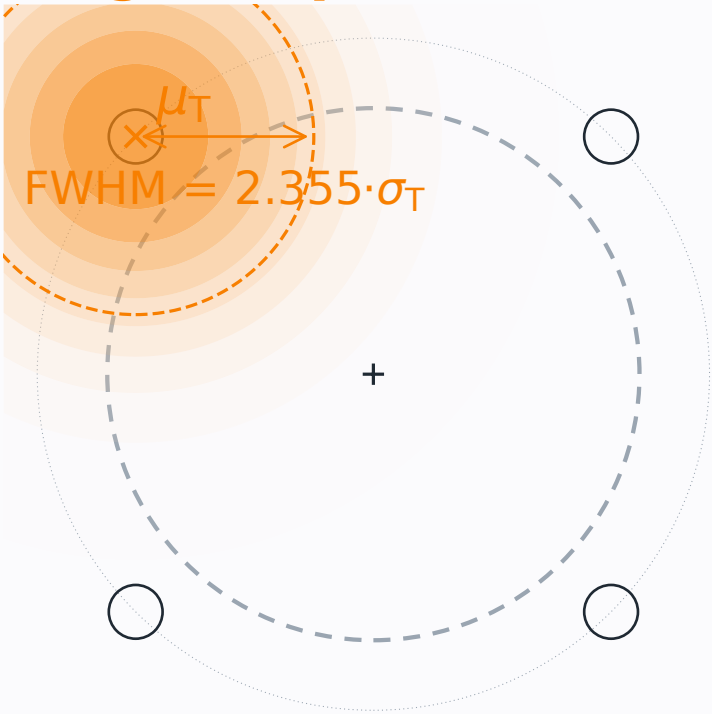
$g_{HP} < 0$ (suppression) · sustained $h(t) = 1$

Dynamic distractor



$g_D < 0$ · transient $d(t)$ · μ_D varies per trial

Target capture



$g_T > 0$ · transient $\tau(t)$ · μ_T varies per trial